FARMER-TO-BUYER BLOCKCHAIN MARKETPLACE

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Use Case Report

submitted by

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Kanuru, Vijayawada-520 007

2026-27

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CERTIFICATE

This is to certify that the Use Case report entitled **"Farmer-to-Buyer Blockchain Marketplace"** that is being submitted by **Divija(23501A0516)**, as part of Assignment-2 for the **Blockchain Technology**(**23CS4701B)** course in **4-1** during the academic year **2026-27**.

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1. INTRODUCTION

1.1. Overview of Blockchain Technology

Blockchain technology is a decentralized and distributed ledger system that records transactions across multiple nodes, ensuring transparency, security, and immutability. Unlike traditional databases, blockchain operates on a peer-to-peer network, eliminating the need for central authorities. Each block in the chain contains a cryptographic hash of the previous block, making data tampering virtually impossible. Originally designed for cryptocurrency transactions, blockchain technology is now used in various domains, including finance, supply chain management, healthcare, and agricultural trading.

1.2. Relevance of Blockchain in Farmer-to-Buyer Marketplace

In traditional agricultural trade, farmers often rely on intermediaries to sell their produce to buyers. These systems face challenges such as delayed payments, lack of transparency, and difficulties in verifying product origin and delivery details. Blockchain technology addresses these issues by recording transactions securely and transparently, ensuring reliable and verifiable trade records.

A blockchain-based farmer-to-buyer marketplace provides a secure mechanism for managing produce listings, orders, and payments. By leveraging smart contracts, transactions and escrow payments can be handled automatically without unnecessary intermediaries, reducing processing delays and human errors. Additionally, supply-chain tracking enables the recording of shipment checkpoints, allowing buyers to verify the movement of agricultural produce throughout the delivery process.

The adoption of blockchain in agricultural trade enhances trust between farmers and buyers, enabling a secure, transparent, and reliable marketplace. This transformation ensures that orders, payments, and delivery records remain traceable, ultimately improving payment security and operational efficiency.

Furthermore, the decentralized nature of blockchain ensures that transaction records are difficult to alter, reducing dependence on a single centralized system. This feature is particularly beneficial when multiple parties—such as farmers, buyers, transporters, and dispute arbitrators—need access to transaction information for verification and dispute resolution. Additionally, the escrow mechanism holds the buyer's payment until delivery is confirmed, while an automatic release mechanism can transfer payment to the farmer after seven days if no response is received. This structured payment process helps reduce payment uncertainty and supports fairer agricultural trade. As digital technologies continue to develop, blockchain-based marketplaces offer a transparent and reliable approach to managing agricultural transactions securely and efficiently.

2. BACKGROUND

Traditional agricultural trading systems face numerous challenges, often leading to inefficiencies, payment delays, and disputes between farmers and buyers. These challenges arise due to the reliance on intermediaries, limited transparency, and manual transaction processing. Below are some key issues with the existing agricultural trading systems:

2.1. Dependence on Intermediaries

Many farmers rely on brokers and agents to sell their agricultural produce to buyers. These intermediaries often reduce the farmer's profit by adding commissions and limiting direct communication between farmers and consumers.

2.2. Delayed Payments and Payment Uncertainty

Farmers frequently face delayed payments after delivering their produce due to the lack of secure payment mechanisms. Uncertainty regarding payment settlement creates financial difficulties for farmers and reduces trust between trading parties.

2.3. Inefficiencies in Order Processing

Agricultural transactions often involve manual order management, payment verification, and delivery confirmation. These procedures can cause delays, errors, and difficulties in maintaining accurate transaction records.

2.4. Lack of Transparency and Traceability

Traditional agricultural trading systems often lack reliable mechanisms for tracking produce from farmers to buyers. This makes it difficult for buyers to verify product origin, shipment details, and delivery status, creating uncertainty throughout the supply chain.

2.5. Challenges in Delivery Verification

Verifying whether agricultural produce has been shipped and delivered as agreed can be difficult when multiple parties are involved. A lack of reliable delivery records can lead to disagreements between farmers, buyers, and transporters.

2.6. Security Risks and Data Tampering

Centralized agricultural trading systems may be vulnerable to unauthorized modifications, data breaches, and manipulation of transaction records. Such risks can affect order details, payment information, and delivery records, leading to losses and disputes.

2.7. Difficulty in Coordinating Multiple Parties

Agricultural trade involves several participants, including farmers, buyers, transporters, and dispute arbitrators. Coordinating these parties through separate systems can create inconsistencies in transaction records and complicate the overall trading process.

2.8. Disputes Between Farmers and Buyers

Farmers and buyers may disagree regarding product delivery, order fulfillment, or payment settlement. Without a structured dispute resolution mechanism, resolving these disagreements can become time-consuming and may result in financial losses for either party.

2.9. Limited Access to Reliable Transaction Information

Many traditional agricultural trading systems may not provide all participants with convenient access to order status, payment details, and shipment updates. This lack of accessible information can cause confusion, repeated communication, and delays in decision-making.

2.10. Lack of Automated Payment Settlement

Farmers often depend on buyers to confirm delivery and release payments manually. Without automated settlement mechanisms, payments may be delayed even after delivery. A secure escrow mechanism can hold the buyer's payment and release it according to predefined smart contract conditions.

Due to these challenges, there is a growing need for a secure, transparent, and automated agricultural trading system. Blockchain technology offers a promising solution by recording transactions transparently, enabling supply-chain tracking, and using smart contracts to manage escrow payments and support structured dispute resolution.

3. BLOCKCHAIN BASICS

Blockchain is a distributed ledger technology that records transactions securely and transparently. It allows participants to interact and exchange information without relying entirely on a central authority. The records are linked using cryptographic techniques, making unauthorized modifications difficult to perform without detection. In the **Farmer-to-Buyer Blockchain Marketplace with Escrow**, blockchain concepts and smart contracts support transparent agricultural transactions, secure payment handling, and supply chain tracking.

3.1. Decentralization

Decentralization is one of the key characteristics of blockchain technology. Instead of depending on a single central authority, the network distributes the responsibility of maintaining and validating records among participating nodes. This reduces reliance on a single point of control and improves transparency. In an agricultural marketplace, decentralization can help create a more transparent interaction between farmers and buyers by recording transactions and enforcing predefined rules through smart contracts.

3.2. Immutability

Immutability refers to the difficulty of altering transaction records once they have been recorded on a blockchain. Each block is linked to previous blocks through cryptographic hashes, helping protect the integrity of the stored information. This characteristic supports trust and accountability because unauthorized changes can be detected. In the Farmer-to-Buyer Blockchain Marketplace with Escrow, blockchain-based records can support the traceability of agricultural transactions, order status changes, and supply chain checkpoints.

3.3. Transparency

Transparency allows participants to verify relevant transactions and records according to the permissions and visibility provided by the blockchain system. Shared records can improve accountability and reduce information gaps between participants. In an agricultural marketplace, transparent transaction records can help farmers and buyers track order progress, verify payment-related actions, and review recorded supply chain checkpoints.

3.4. Smart Contracts

Smart contracts are programs deployed on a blockchain that automatically execute predefined rules when specified conditions are met. They reduce the need for manual intervention in processes such as transaction validation, payment handling, and status updates. In the Farmer-to-Buyer Blockchain Marketplace with Escrow, smart contracts manage crop listings, order placement, escrow payments, delivery confirmation, and dispute resolution. The escrow mechanism holds the buyer's payment until the required conditions are satisfied, while predefined rules support refunds and payment settlement.

3.5. Consensus Mechanisms

Consensus mechanisms are protocols used by blockchain networks to agree on the validity and ordering of transactions. Common mechanisms include Proof of Work (PoW), Proof of Stake (PoS), and Delegated Proof of Stake (DPoS). Each mechanism uses a different approach to validate transactions and maintain agreement across the network. The choice of consensus mechanism affects factors such as security, performance, and energy consumption. The concepts are relevant to understanding blockchain platforms, although this project's demonstration uses a local Hardhat development network rather than a deployed public blockchain.

3.6. Cryptographic Security

Cryptographic techniques help protect blockchain data and verify the authenticity and integrity of transactions. Hash functions generate fixed-length representations of data, while digital signatures can help verify the identity of transaction signers. These techniques contribute to the security and reliability of blockchain applications. In the Farmer-to-Buyer Blockchain Marketplace with Escrow, cryptographic methods and smart contract access controls support secure transaction processing, role-based operations, and the integrity of recorded supply chain information.

Blockchain technology provides a foundation for building transparent and reliable digital applications. Its characteristics, including decentralization, immutability, transparency, smart contracts, and cryptographic security, are relevant to agricultural marketplaces that aim to improve trust between farmers and buyers.

In the proposed Farmer-to-Buyer Blockchain Marketplace with Escrow, smart contracts define the rules for crop listings, order processing, payment escrow, delivery confirmation, and dispute resolution. Supply chain checkpoints provide additional information about the movement of agricultural products. Together, these features demonstrate how blockchain concepts can support transparent agricultural transactions and accountable payment handling

4. USE CASE OVERVIEW

The use case for a **Farmer-to-Buyer Blockchain Marketplace with Escrow** aims to improve agricultural transactions by leveraging blockchain technology and smart contracts. This system supports transparent crop listings, secure escrow-based payments, supply chain tracking, and dispute resolution, helping build trust between farmers and buyers.

4.1. Objectives

The primary objective of the **Farmer-to-Buyer Blockchain Marketplace with Escrow** is to provide a transparent digital platform for agricultural transactions between farmers and buyers. Traditional agricultural trading may involve intermediaries, payment delays, and difficulties in tracking orders. By leveraging blockchain concepts and smart contracts, the system supports crop listing, order management, and secure payment handling through an escrow mechanism.

A key challenge in agricultural marketplaces is building trust between farmers and buyers. The proposed system uses smart contracts to enforce predefined transaction rules and maintain consistent order status updates. Farmer and buyer verification helps restrict marketplace operations to verified participants, while the escrow mechanism holds the buyer's payment until the required conditions for settlement are met.

Another important goal is to provide secure and reliable payment settlement. When a buyer places an order, the payment is locked in escrow using a mock stablecoin in the demonstration. After the buyer confirms delivery, the smart contract calculates the platform fee and transfers the remaining payment to the farmer. If an eligible order is cancelled before shipment, the system supports a refund and restores the available crop quantity.

The system also supports dispute resolution through smart contracts. If a buyer raises a dispute after shipment, an authorized arbiter can decide how the payment should be divided between the buyer and the farmer. This predefined process provides a structured way to handle disagreements and reduces the need for informal payment arrangements.

Transparency is another important benefit of the proposed marketplace. Crop details, order status changes, payment-related actions, and supply chain checkpoints can be recorded and reviewed within the system. Verified transporters can submit checkpoints containing location, temperature, and additional notes, helping participants review the recorded movement information of agricultural products.

By automating transaction rules, payment settlement, refunds, and dispute handling, the system aims to reduce manual coordination between farmers and buyers. These features can make agricultural trading more organized and improve accountability throughout the transaction process.Security is another key consideration. The smart contracts use access controls and protections against certain unsafe transaction patterns to restrict authorized operations and help protect escrow payments. Overall, the proposed system demonstrates how blockchain concepts and smart contracts can support transparent agricultural trading, payment security, and supply chain traceability.

4.2. Scope of the System

The Farmer-to-Buyer Blockchain Marketplace with Escrow aims to provide a secure, transparent, and decentralized platform for agricultural trading. The scope of the system includes the following features:

1. **Farmer Registration and Crop Listing:** Enables farmers to register, add crop details, specify prices, and manage available quantities directly on the marketplace.
2. **Buyer Registration and Crop Purchasing:** Allows buyers to browse available crops, view product details, compare prices, and place orders directly with farmers.
3. **Smart Contract-Based Escrow:** Locks the buyer's payment in escrow until the required conditions are satisfied, helping ensure secure and transparent transactions.
4. **Blockchain-Based Transaction Tracking:** Records important transaction details and order status updates on the blockchain, providing transparency and traceability.
5. **Secure Payment and Refund Management:** Supports payment settlement, automatic platform fee calculation, and eligible refunds through smart contract logic. A mock stablecoin is used for payment demonstration.
6. **Order and Delivery Management:** Enables users to track order progress and allows buyers to confirm delivery before the escrow payment is released to the farmer.
7. **Dispute Resolution:** Supports dispute reporting and resolution through predefined smart contract rules and authorized dispute handling.
8. **Fraud Prevention and Trust:** Uses participant verification, blockchain records, and smart contract validation to reduce unauthorized activities and improve trust between farmers and buyers.

4.3. System Architecture

The architecture of the Farmer-to-Buyer Blockchain Marketplace with Escrow is illustrated in **Figure 4.1** and consists of the following components.

4.3.1. User Layer

This layer consists of the primary users who interact with the Farmer-to-Buyer Blockchain Marketplace with Escrow. It defines the roles and responsibilities of different participants involved in agricultural transactions. The main participants are farmers, buyers, transporters, and dispute resolvers. Each participant performs specific activities based on their role and permissions within the system. Farmers list their agricultural produce, while buyers place orders and make payments through the escrow mechanism. Transporters record shipment checkpoints, and dispute resolvers handle transaction disputes according to predefined smart contract rules.

Farmers:

- **Registration and Profile Management:** Register on the platform and maintain their profile information to access marketplace services.
- **Crop Listing:** Add crop details, including crop name, price, available quantity, and other relevant product information.
- **Listing Management:** Update crop details, modify prices, and manage the availability of listed products.
- **Order Management:** View incoming orders and monitor their status throughout the transaction process.
- **Payment Tracking:** Check payment status and transaction details associated with their crop sales.

Buyers:

- **Crop Browsing:** Browse available crops and view product details, prices, and quantities before purchasing.
- **Product Selection:** Compare available products and select crops according to their requirements.
- **Order Placement:** Place orders directly through the marketplace, reducing the need for intermediaries.
- **Secure Payment:** Initiate payments through the smart contract-based escrow mechanism to support secure transactions.
- **Order Tracking:** Monitor order status and access relevant transaction information.
- **Delivery Confirmation:** Confirm receipt of products before payment release, if supported by the implemented smart contract.

Administrator / Dispute Resolver:

- **Platform Monitoring:** Monitor marketplace activities and review relevant transaction information, where administrative features are available.
- **Transaction Verification:** Examine transaction details to support transparency and accountability.
- **Dispute Handling:** Assist in resolving disputes between farmers and buyers according to predefined rules, if implemented.
- **Fair Transaction Management:** Help maintain trust between participants by following the platform's defined procedures for handling transaction-related issues.

4.3.2. Blockchain Layer

This layer contains the Solidity smart contracts that implement the marketplace's core business rules.

Smart Contracts:

- **UserRegistry:** Manages participant registration and verification.
- **MockStablecoin:** Provides test tokens for simulating payment transactions.
- **Marketplace:** Manages crop listings, orders, escrow payments, cancellations, payment settlement, and dispute resolution.
- **SupplyChainTracker:** Records shipment checkpoints submitted by verified transporters.

Transaction Processing:

When a buyer places an order, the Marketplace contract validates the request and locks the payment in escrow. When the required conditions are satisfied, the contract settles the payment and applies the configured platform fee. Eligible cancellations and disputes are handled according to predefined contract rules.

Local Blockchain Network:

The contracts are deployed and tested on a local Hardhat network. This environment allows transaction execution and contract behavior to be demonstrated without claiming deployment to a public blockchain.

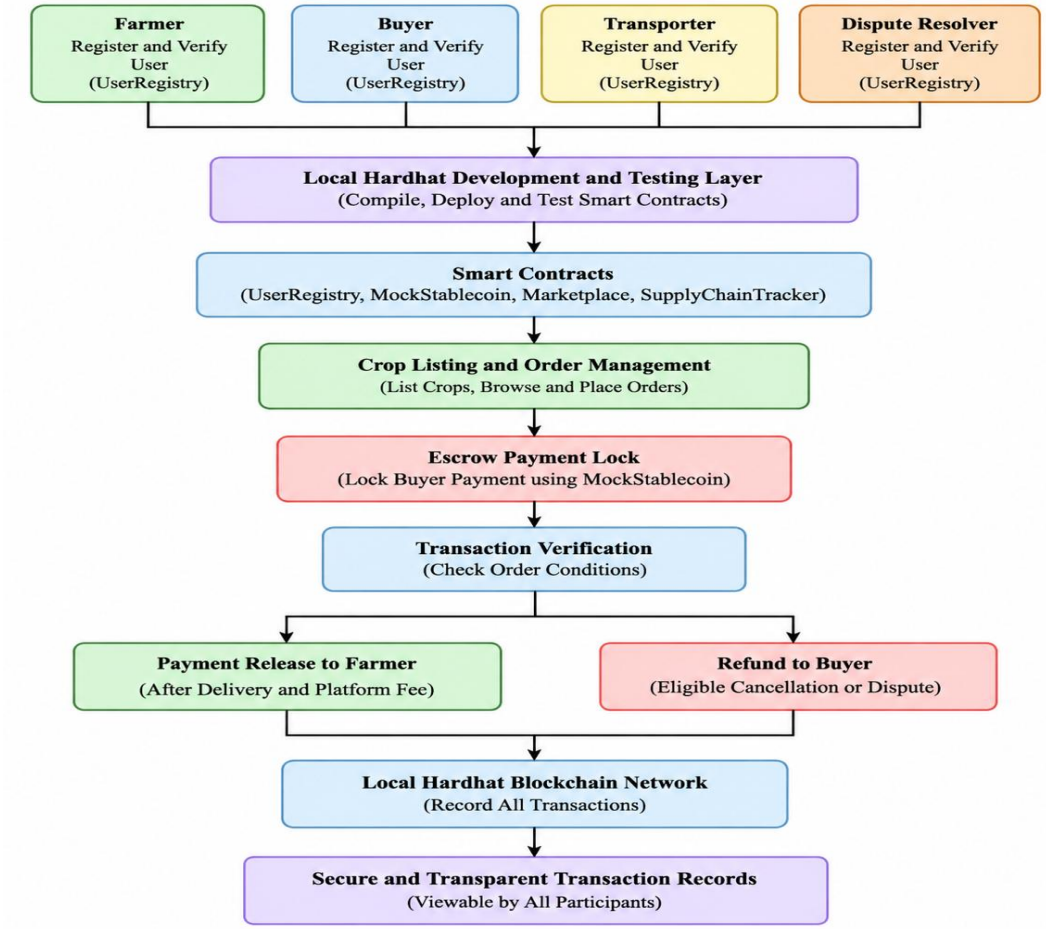


Figure 4.1: System Architecterr of Farmer-to-Buyer Blockchain Marketplace with Escrow

5. IMPLEMENTATION

5.1. Setting Up the Blockchain Environment

To implement the Farmer-to-Buyer Blockchain Marketplace with Escrow, the development environment was configured using Solidity and the Hardhat development framework. Four smart contracts, namely UserRegistry, MockStablecoin, Marketplace, and SupplyChainTracker, were developed to manage user registration, crop listings, escrow payments, order processing, dispute resolution, and supply chain checkpoints. The contracts were compiled, deployed, and tested on a local Hardhat network. The MockStablecoin was used to simulate payments during testing rather than process real monetary transactions. The smart contracts were tested to verify access control, payment settlement, refunds, order status transitions, and escrow functionality according to predefined rules.

5.2. Defining Smart Contracts

5.2.1. Writing the Smart Contract for Agricultural Marketplace

The smart contract is responsible for managing crop listings, processing orders, and handling escrow-based payments in the Farmer-to-Buyer Blockchain Marketplace. It defines the rules for placing orders, locking buyer payments, releasing funds to farmers, and processing eligible refunds. The contract maintains relevant transaction details, including buyer and farmer addresses, crop information, order amounts, and payment status. It also calculates the platform fee according to the predefined rules. By executing these operations through smart contracts, the system improves transaction transparency, reduces manual intervention, and ensures that payments are processed according to the conditions defined in the contract.

```
struct Crop {
    uint256 id;
    address farmer;
    string cropName;
    uint256 price;
    uint256 quantity;
    bool isAvailable;
}
struct Order {
    uint256 id;
    address buyer;
    address farmer;
    uint256 cropId;
    uint256 amount;
    bool isPaid;
    bool isCompleted;
}
```

5.2.2. Registering Farmers and Buyers

The UserRegistry smart contract manages participant registration and verification for the marketplace. Farmers and buyers register their blockchain addresses and complete the required verification process before performing restricted marketplace operations. Verified farmers can create crop listings, while verified buyers can place orders according to the Marketplace contract's rules. Access-control checks help prevent unauthorized participants from executing restricted operations and support reliable transaction processing.

```
mapping(address => bool) internal registeredUsers;
function registerUser() external {
    require(!registeredUsers[msg.sender],"Already registered");
    registeredUsers[msg.sender] = true;
}
```

5.2.3. Listing Crops and Placing Orders

Once registered, farmers can list their crops on the marketplace by specifying the crop name, price, and available quantity. Each crop is assigned a unique identifier to enable easy tracking and management. Buyers can browse the available listings, select the required crops, and place orders through the web application. The smart contract manages the order details and, when a buyer initiates a purchase, locks the payment in escrow according to the contract rules. This process improves transparency, supports secure transactions, and helps ensure that payments are handled according to predefined conditions.

```
function listCrop(
    string memory cropName,
    uint256 price,
    uint256 quantity
) external {
    require(bytes(cropName).length > 0, "Invalid crop name");
    require(price > 0, "Invalid price");
    require(quantity > 0, "Invalid quantity");
    uint256 cropId = nextCropId++;
    crops[cropId] = Crop({
        id: cropId,
        farmer: msg.sender,
        cropName: cropName,
        price: price,
        quantity: quantity,
        isAvailable: true
    });
    farmerToCrops[msg.sender].push(cropId);
}
```

5.2.4. Viewing Crop Listings Available in the Marketplace

The Marketplace smart contract maintains crop listings created by registered and verified farmers. Each listing contains relevant details such as crop type, available quantity, price, harvest date, origin, and metadata hash. Buyers can retrieve the available listing information through the smart contract's supported functions. The stored details help identify agricultural produce and review its availability before placing an order. This functionality supports transparent crop listing management and provides the information required for agricultural transactions.

```
function viewCrops() external view returns (Crop[] memory) {
    uint256[] memory cropIds = farmerToCrops[msg.sender];
    Crop[] memory listedCrops = new Crop[](cropIds.length);

    for (uint256 i = 0; i < cropIds.length; i++) {
        listedCrops[i] = crops[cropIds[i]];
    }

    return listedCrops;
}
```

5.2.5. Verifying Crop Availability and Order Validity

Before processing an order, the Marketplace smart contract validates the requested crop quantity and checks whether sufficient stock is available. The requested quantity must be valid and must not exceed the available quantity. If the order satisfies the contract's validation rules, the transaction can proceed and the corresponding quantity is accounted for. These checks help prevent invalid orders and over-purchasing, supporting consistent crop inventory and reliable order processing.

```
function verifyCropAvailability(uint256 cropId){
    return crops[cropId].isAvailable
        && crops[cropId].quantity > 0;
}
function verifyOrderValidity(uint256 orderId){
    Order memory order = orders[orderId];
    return order.id != 0
        && order.buyer == msg.sender
        && !order.isCompleted;
}
```

5.2.6. Updating Crop Listings and Managing Orders

The Marketplace smart contract manages crop listings and order status throughout the transaction process. Verified farmers can create crop listings with the required agricultural details, while buyers can place orders according to the available quantity and predefined contract rules. The contract tracks order progress through states such as Placed, Shipped, Delivered, Completed, Cancelled, Disputed, and Resolved. Access-control checks restrict sensitive operations to authorized participants. This approach supports consistent order management and transparent transaction processing.

```
  function updateCropQuantity(
      uint256 cropId,
      uint256 newQuantity
  ) external {
      require(
          crops[cropId].farmer == msg.sender,
          "Not the crop owner"
      );

      crops[cropId].quantity = newQuantity;
      crops[cropId].isAvailable = newQuantity > 0;
  }
function updateOrderStatus(
    uint256 orderId,
    bool completed
) external {
    require(orders[orderId].id != 0, "Order does not exist");
    require(
        orders[orderId].farmer == msg.sender,
        "Not the farmer"
    );

    orders[orderId].isCompleted = completed;
}
```

5.3. Deploying and Testing Smart Contracts

After development, the smart contracts were compiled and deployed in a local Hardhat development environment. The deployment and demonstration scripts were used to execute and verify important marketplace operations. The four contracts work together to manage participant registration, mock token payments, crop listings, escrow transactions, dispute resolution, and supply chain checkpoints. Automated tests were used to check successful transactions, invalid operations, cancellation and refunds, payment settlement, access restrictions, and dispute handling. This local testing environment demonstrates the smart contracts' functionality without requiring deployment to a public blockchain.

5.4. Workflow:

The workflow of the Farmer-to-Buyer Blockchain Marketplace with Escrow describes the sequence of smart contract operations involved in crop listing, order placement, escrow payment handling, delivery confirmation, and dispute resolution.

The process begins when a farmer registers and completes the required verification before creating a crop listing. A buyer selects an available listing and places an order. The Marketplace smart contract validates the requested quantity and locks the buyer's payment in escrow using the MockStablecoin.

After shipment, the buyer can confirm delivery, allowing the contract to settle the payment after calculating the applicable platform fee. If an eligible cancellation occurs before shipment, the contract refunds the buyer and restores the available crop quantity. If a dispute is raised after shipment, an authorized arbiter determines the refund percentage according to the contract rules. Verified transporters can also submit supply chain checkpoints through the SupplyChainTracker contract. These operations are demonstrated and tested on the local Hardhat network.

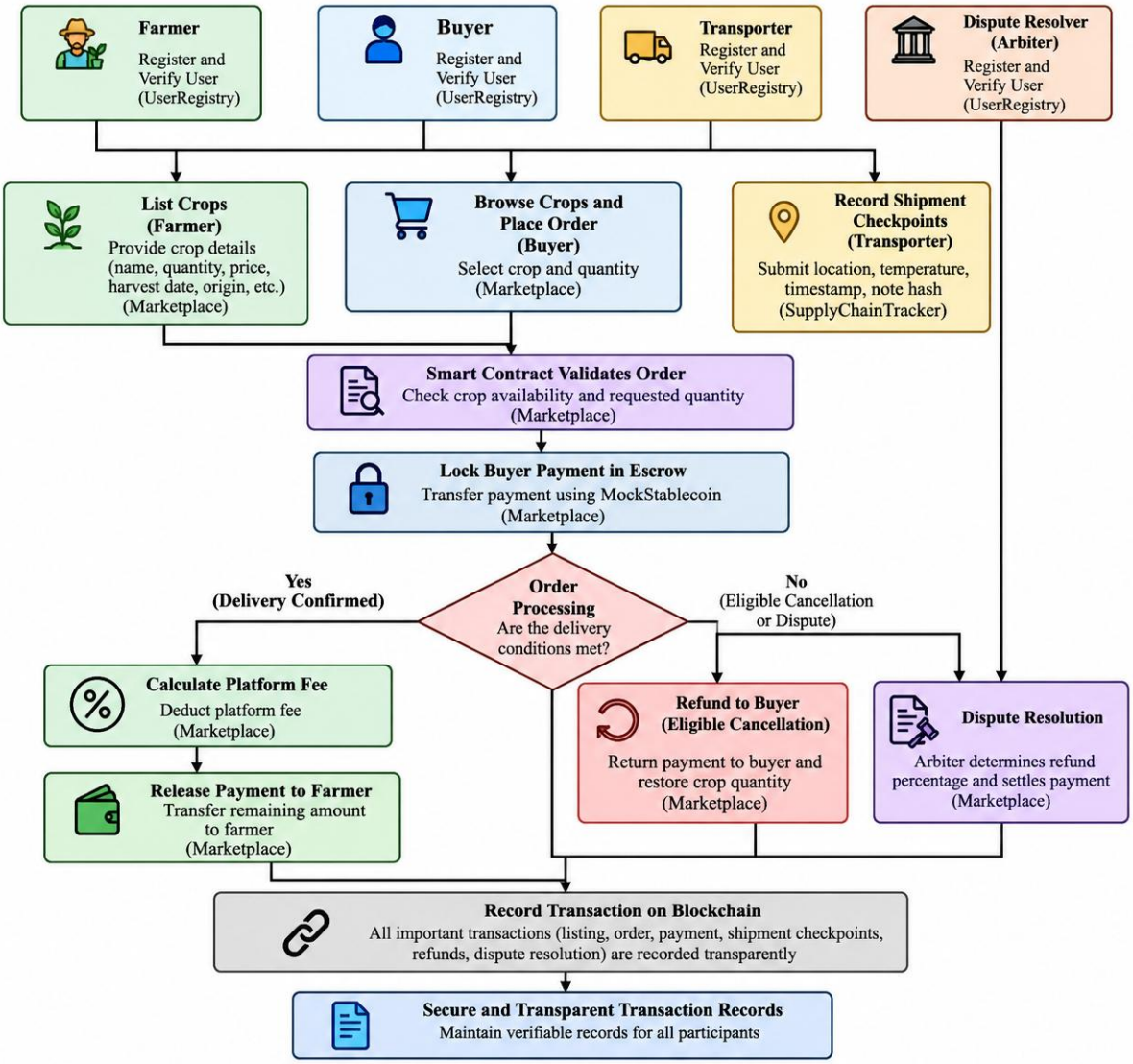


Figure 5.1: Workflow of Farmer-to-Buyer Blockchain Marketplace with Escrow

6. BENEFITS

Blockchain technology improves agricultural trading by providing transparency, secure transactions, and automated payment processing. Traditional agricultural marketplaces may involve intermediaries, payment delays, and difficulties in tracking orders. By integrating blockchain and smart contracts, the Farmer-to-Buyer Marketplace with Escrow helps improve trust, reduce payment risks, and simplify transactions. The key benefits are described below.

6.1. Enhanced Security and Immutability

Blockchain maintains verifiable transaction records that are resistant to unauthorized modification. This helps farmers and buyers track transactions and verify payment details, improving transparency and reducing the risk of fraudulent activities.

6.2. Direct Farmer to Buyer Transactions

The marketplace enables farmers to list crops and sell their products directly to buyers. This reduces dependence on intermediaries, improves market accessibility, and helps farmers reach potential customers more efficiently.

6.3. Secure Escrow Based Payments

The escrow mechanism holds the buyer's payment until the predefined transaction conditions are satisfied. This helps protect both parties by ensuring that funds are released according to the smart contract rules.

6.4. Automated Payment Processing

Smart contracts automate payment-related operations, including payment release and eligible refunds. They also support platform fee calculation according to predefined rules, reducing manual intervention and improving transaction efficiency.

6.5. Improved Crop and Order Management

Farmers can manage crop listings, prices, and available quantities, while buyers can browse products and place orders through the marketplace. This simplifies agricultural trading and helps users access relevant crop and order information.

6.6. Reduced Transaction Disputes

Transparent transaction records and predefined smart contract rules help reduce misunderstandings between farmers and buyers. Access to transaction details improves accountability and supports dispute investigation when required.

6.7. Increased Trust and Reliability

Blockchain-based records and escrow payment handling help build trust between farmers and buyers. Participants can verify supported transactions, while automated contract rules promote consistent payment processing.

6.8. Cost and Time Efficiency

The digital marketplace simplifies crop discovery, order placement, and payment handling. Reducing manual processes and unnecessary intermediaries can save time and potentially lower transaction-related costs.

6.9. Traceability of Transactions

Each recorded blockchain transaction can be verified using its transaction details. This provides a traceable history of supported activities, helping participants review order and payment-related operations.

6.10. Improved Accessibility and Digital Trading

The proposed digital marketplace provides a structured approach for farmers and buyers to participate in agricultural trading. It helps users access crop information and conduct supported transactions without relying entirely on traditional trading methods.

In conclusion, the Farmer-to-Buyer Blockchain Marketplace with Escrow combines blockchain technology and smart contracts to improve agricultural trading. Its major benefits include transparent transactions, secure escrow payments, automated payment processing, and simplified crop and order management. These features can strengthen trust between farmers and buyers while making agricultural transactions more efficient.

7. CHALLENGES

Although the Farmer-to-Buyer Blockchain Marketplace with Escrow offers several benefits, its implementation involves certain technical and operational challenges. These limitations must be considered to ensure secure, reliable, and efficient agricultural transactions.

7.1. Scalability Issues

Blockchain networks may experience slow transaction processing and increased gas fees when transaction volumes are high. As the number of farmers, buyers, and orders increases, the system may require suitable blockchain infrastructure to maintain performance.

7.2. Smart Contract Security

Smart contracts handle important operations such as escrow payments, payment releases, and refunds. Errors or vulnerabilities in contract logic may lead to incorrect transactions or financial losses. Therefore, careful testing and security audits are important before deployment.

7.3. Lack of User Awareness and Adoption

Some farmers and buyers may have limited knowledge of blockchain technology, digital wallets, and cryptocurrency-based transactions. A simple user interface and clear instructions are necessary to encourage adoption and make the marketplace accessible.

7.4. Dependence on Internet Connectivity

The marketplace requires internet connectivity to access crop listings, place orders, and interact with blockchain services. Users in areas with poor network connectivity may experience difficulties accessing the platform or completing transactions.

7.5. Payment and Escrow Management

Managing escrow payments requires accurate transaction validation and clearly defined conditions for payment release and refunds. Incorrect contract logic or failed transactions may delay settlement and create difficulties for farmers and buyers.

Despite these challenges, the proposed marketplace can improve agricultural trading through transparent records, direct farmer-to-buyer transactions, and smart contract-based escrow payments. Proper testing, secure contract development, user-friendly design, and reliable blockchain infrastructure can help address these limitations.

8. CONCLUSION

The report explores the integration of blockchain technology into agricultural trading to address challenges such as intermediary dependency, payment delays, lack of transparency, and transaction-related disputes. The proposed Farmer-to-Buyer Blockchain Marketplace with Escrow provides a digital platform that connects farmers and buyers directly. By using blockchain technology and smart contracts, the system aims to improve transaction transparency, secure payment handling, and trust between participants.

The marketplace enables farmers to list crops with prices and available quantities, while buyers can browse products and place orders. The escrow mechanism holds the buyer's payment until the predefined conditions are satisfied, supporting payment release to farmers and eligible refunds. Blockchain transaction records provide traceability and help maintain data integrity. Overall, the proposed system offers a structured approach to making agricultural trading more transparent, reliable, and efficient.

8.1. Future Outlook for Enhancements

To further improve the system, artificial intelligence (AI) can be integrated to analyze crop prices, market demand, and historical trading patterns. This would help farmers make informed decisions about crop pricing and assist buyers in identifying suitable products. AI-based analytics could also help detect unusual transaction patterns and identify potentially fraudulent activities.

Integrating decentralized identity (DID) solutions could improve user authentication and privacy by enabling secure verification of farmers and buyers. Additionally, interoperability with multiple blockchain networks could support broader adoption and enable transactions across different blockchain platforms.

Scalability solutions, such as Layer 2 protocols, could be introduced to reduce transaction costs and improve processing speed as the number of users increases. Integrating real payment gateways, logistics tracking, and advanced dispute-resolution mechanisms could further improve the usability of the marketplace.

By incorporating these enhancements, the Farmer-to-Buyer Blockchain Marketplace with Escrow could become a more scalable, secure, and efficient platform that supports transparent agricultural trading and strengthens trust between farmers and buyers.

9. SDG's ADDRESSED

The Farmer-to-Buyer Blockchain Marketplace with Escrow aligns with selected United Nations Sustainable Development Goals (SDGs) by promoting agricultural market access, encouraging digital innovation, and supporting transparent transactions. The key SDGs addressed by the proposed system and their justifications are described below:

SDG 9: Industry, Innovation, and Infrastructure

The implementation of blockchain technology in the Farmer-to-Buyer Blockchain Marketplace with Escrow promotes digital innovation in agricultural trading. By using decentralized and tamper-resistant transaction records, the system improves transparency and reliability. Smart contracts automate escrow payments and payment settlement according to predefined conditions, reducing manual intervention and supporting efficient transactions. This contributes to the development of innovative digital infrastructure for agricultural commerce.

SDG 12: Responsible Consumption and Production

The Farmer-to-Buyer Blockchain Marketplace with Escrow supports responsible agricultural trading by providing clear information about crop availability, prices, and quantities. Direct communication between farmers and buyers can improve market access and help users make informed purchasing decisions. Transparent transaction records also improve accountability throughout the trading process. Future integration of inventory management and demand forecasting could further help reduce food waste and improve resource utilization.

SDG 13: Climate Action

The proposed marketplace can indirectly support climate action by improving digital agricultural trading and reducing dependence on certain paper-based administrative processes. By enabling farmers and buyers to access crop information and manage transactions online, the system may reduce some unnecessary travel and paperwork. However, the actual environmental impact depends on user adoption, transportation practices, and the energy consumption of the blockchain network.

By integrating blockchain technology and smart contracts, the Farmer-to-Buyer Blockchain Marketplace with Escrow supports digital innovation, transparent agricultural transactions, and more responsible trading practices. These objectives align with the selected Sustainable Development Goals and provide opportunities for future improvements in agricultural sustainability.

10. REFERENCES

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11. APPENDIX A

https://drive.google.com/drive/folders/1C5gcReByVPbKwwHksh5yk8br9jRgOLX_?usp=drive_link